Chapter 7. Function Spaces

The space C(X), Arzelà–Ascoli, Weierstrass approximation, sequence spaces, improper integrals, spaces of integrable functions

Contents

7.1 The space $C(X)$ **2**

7.2 Sequence spaces **3**

7.3 Improper integrals **3**

7.4 Spaces of integrable functions **4**

7.5 Selected exercises **6**

7.1 The space $C(X)$

Motivation. $C(X)$ with the sup norm is complete, and polynomials are dense in it.

A **norm** on a real or complex vector space V is a function $\|\cdot\|$ with $\|v\| \geq 0$ (equality iff $v = 0$), $\|\lambda v\| = |\lambda|\|v\|$, and $\|u + v\| \leq \|u\| + \|v\|$; V is then a **normed space**, with distance $\|u - v\|$, and convergence, Cauchy sequences, open and closed sets are defined as in $\mathbb{R}^n$. A normed space in which every Cauchy sequence converges is **complete** (a **Banach space**). For compact $X \subset \mathbb{R}^n$, $C(X)$ is the space of continuous $f : X \to \mathbb{C}$ with $\|f\|_{\sup} = \max_X |f|$; convergence in this norm is uniform convergence.

Theorem 7.1 (7.1.1)

For compact X, every Cauchy sequence in $C(X)$ converges: $C(X)$ is complete.

Proof. A Cauchy sequence in $C(X)$ is uniformly Cauchy, so converges uniformly (Theorem 6.2) to a function that is continuous (Theorem 6.1). ■

Heine–Borel fails in $C(X)$: $\{x^n\}$ on $[0, 1]$ is closed and bounded, yet no subsequence converges uniformly. The missing ingredient is equicontinuity. A family $\mathcal{F} \subset C(X)$ is **equicontinuous** if for every $\epsilon > 0$ there is $\delta > 0$ with $|f(x) - f(y)| < \epsilon$ for all $f \in \mathcal{F}$ and $\|x - y\| < \delta$; it is **bounded** if $\sup_{f \in \mathcal{F}} \|f\|_{\sup} < \infty$.

Proposition 7.2 (7.1.2)

If X is compact and (f_n) converges uniformly in $C(X)$, then $\{f_n\}$ is equicontinuous.

Proof. Choose N with $\|f_n - f_N\|_{\sup} < \epsilon/3$ for $n \geq N$ (Theorem 6.2) and δ working with $\epsilon/3$ for the uniformly continuous $f_1, \dots, f_N$ (Theorem 3.18 of Volume I). For $n > N$, $\|x - y\| < \delta$, $|f_n(x) - f_n(y)| \leq |f_n(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f_n(y)| < \epsilon$. ■

Theorem 7.3 (7.1.3, Arzelà–Ascoli)

If X is compact and $\mathcal{F} \subset C(X)$ is bounded and equicontinuous, every sequence in $\mathcal{F}$ has a uniformly convergent subsequence.

Proof. Let $f_n \in \mathcal{F}$, $|f_n| \leq M$. X has a countable dense $E = \{p_1, p_2, \dots\}$ (centres of finitely many $1/k$-balls covering X, all k).

Choose nested subsequences $(f_{k,n})_n$ with $(f_{k,n}(p_k))_n$ convergent; the diagonal $g_n = f_{n,n}$ is eventually a subsequence of each, so $(g_n(p_k))$ converges for every k.

Take δ from equicontinuity for $\epsilon/3$, points $p_{k_1}, \dots, p_{k_r}$ forming a δ-net, and N with $|g_m(p_{k_j}) - g_n(p_{k_j})| < \epsilon/3$ for $m, n \geq N$, all j. For $x \in X$ and $p = p_{k_j}$ with $\|x - p\| < \delta$, $|g_m(x) - g_n(x)| \leq |g_m(x) - g_m(p)| + |g_m(p) - g_n(p)| + |g_n(p) - g_n(x)| < \epsilon$. So (g_n) converges in $C(X)$ (Theorem 7.1). ■

Combined with Proposition 7.2: a subset of $C(X)$ has compact closure iff it is bounded and equicontinuous.

Theorem 7.4 (7.1.4, Weierstrass approximation)

For every continuous $f : [a, b] \to \mathbb{R}$ there are polynomials $P_n \to f$ uniformly on $[a, b]$.

Proof. Take $[a,b]=[0,1]$ and (subtracting $f(0)+x(f(1)-f(0))$) $f(0)=f(1)=0$; extended by 0, f is uniformly continuous on $\mathbb{R}$. Let $Q_n(x)=c_n(1-x^2)^n$ on $[-1,1]$ with $\int_{-1}^1 Q_n=1$. As $(1-x^2)^n\geq 1-nx^2$,

$$\int_{-1}^1(1-x^2)^n dx\geq 2\int_0^{1/\sqrt{n}}(1-nx^2)dx=\frac{4}{3\sqrt{n}}>\frac{1}{\sqrt{n}},$$

so $c_n<\sqrt{n}$, and for $\delta\leq|x|\leq 1$, $Q_n(x)\leq\sqrt{n}(1-\delta^2)^n\to 0$ uniformly. Let

$$P_n(x)=\int_{-1}^1 f(x+t)Q_n(t)\,dt=\int_0^1 f(t)Q_n(t-x)\,dt\qquad(0\leq x\leq 1),$$

a polynomial in x. With $|f(y)-f(x)|<\epsilon/2$ for $|y-x|<\delta$ and $M=\max|f|$,

$$|P_n(x)-f(x)|\leq\int_{-1}^1|f(x+t)-f(x)|Q_n(t)\,dt\leq 2M\cdot 2\sqrt{n}(1-\delta^2)^n+\frac{\epsilon}{2}<\epsilon$$

for large n, uniformly in $x\in[0,1]$. ■

For complex-valued f, approximate real and imaginary parts. Consequently $C[a,b]$ has a countable dense set (polynomials with rational coefficients).

7.2 Sequence spaces

Motivation. ℓ^1, ℓ^2: the simplest infinite-dimensional complete normed spaces, models for Chapter 10.

$\ell^1=\ell^1(\mathbb{N})$ is the space of sequences $a=(a(n))$ with $\|a\|_1=\sum|a(n)|<\infty$, and ℓ^2 those with $\|a\|_2=(\sum|a(n)|^2)^{1/2}<\infty$. ℓ^2 is a vector space with inner product $\langle a,b\rangle=\sum a(n)\overline{b(n)}$, which converges absolutely since $|a(n)b(n)|\leq\frac{1}{2}(|a(n)|^2+|b(n)|^2)$; $\|\cdot\|_2$ is the associated norm (Theorem 6.17).

Theorem 7.5 (7.2.1)

ℓ^1 is complete.

Proof. For Cauchy (a_m), $|a_m(n)-a_k(n)|\leq\|a_m-a_k\|_1$, so $a_m(n)\to a(n)$. If $\|a_m-a_k\|_1<\epsilon$ for $m,k\geq N$, then $\sum_{n\leq L}|a_m(n)-a_k(n)|<\epsilon$; letting $k\to\infty$, then $L\to\infty$, $\|a_m-a\|_1\leq\epsilon$ $(m\geq N)$. So $a=a_m-(a_m-a)\in\ell^1$ and $a_m\to a$. ■

Theorem 7.6 (7.2.2)

ℓ^2 is complete.

Proof. As in Theorem 7.5: the coordinates converge, and $\sum_{n\leq L}|a_m(n)-a_k(n)|^2<\epsilon^2$ passes to the limit in k and then in L. ■

Likewise ℓ^∞ (sup norm) is complete. The unit vectors $e_n\in\ell^2$ have $\|e_m-e_n\|_2=\sqrt{2}$: closed bounded sets in infinite dimensions need not be compact.

7.3 Improper integrals

Motivation. Improper integrals handle infinite intervals and unbounded integrands, with tests parallel to series.

Let I be an interval and $f : I \to \mathbb{C}$ Riemann integrable on every compact subinterval. If $I = [a, b)$ ($b \leq \infty$), the **improper integral** $\int_I f = \lim_{c \to b^-} \int_a^c f$ when the limit exists; similarly at a left end point, and for open intervals both ends are treated separately. We write $\int_I |f| < \infty$ when this improper integral converges (**absolute convergence**).

Proposition 7.7 (7.3.1)

If $0 \leq f \leq g$ on I and $\int_I g$ converges, then $\int_I f$ converges.

Proof. $c \mapsto \int_a^c f$ is increasing and bounded by $\int_I g$, so it has a limit as $c \to b^-$ (as for monotone sequences). ■

Proposition 7.8 (7.3.2, comparison test)

If $|f| \leq g$ and $\int_I g$ converges, then $\int_I |f|$ and $\int_I f$ converge and $|\int_I f| \leq \int_I |f| \leq \int_I g$.

Proof. $\int_I |f|$ converges by Proposition 7.7. Cauchy criterion: for $a < c < c' < b$, $|\int_c^{c'} f| \leq \int_c^{c'} |f| \to 0$ as $c, c' \to b$, so $\lim \int_a^c f$ exists (as in Theorem 2.14 of Volume I). The inequality passes to the limit from finite intervals (Theorem 5.5 of Volume I). ■

Example 7.1

$\int_1^\infty x^{-p} dx < \infty$ iff $p > 1$, and $\int_0^1 x^{-p} dx < \infty$ iff $p < 1$. $\int_0^\infty \frac{\sin x}{x}\, dx$ converges (integrate by parts: $\int_1^c \frac{\sin x}{x} dx = \frac{\cos 1}{1} - \frac{\cos c}{c} - \int_1^c \frac{\cos x}{x^2} dx$) but not absolutely (on $[k\pi, (k+1)\pi]$, $\int |\sin x|/x \geq \frac{2}{(k+1)\pi}$).

Proposition 7.9 (7.3.3, integral test)

If $f : [1, \infty) \to (0, \infty)$ is decreasing, then $\sum f(n)$ and $\int_1^\infty f$ converge or diverge together; indeed $\sum_{n=2}^N f(n) \leq \int_1^N f \leq \sum_{n=1}^{N-1} f(n)$, and if $f \to 0$, then $\sum_{n \leq N} f(n) - \int_1^N f$ converges.

Proof. On $[n, n+1]$, $f(n+1) \leq f \leq f(n)$; integrate and sum. The differences $d_N = \sum_{n \leq N} f(n) - \int_1^N f$ decrease ($d_N - d_{N+1} = \int_N^{N+1} f - f(N+1) \geq 0$) and are $\geq f(N) > 0$, so converge. ■

For $f(x) = 1/x$ this gives Euler's constant $\gamma = \lim(1 + \frac{1}{2} + \cdots + \frac{1}{N} - \log N)$.

Theorem 7.10 (7.3.4)

If $\int_I |f| < \infty$, then for every $\epsilon > 0$ there is $\delta > 0$ such that $\int_x^y |f| < \epsilon$ whenever $x < y$ in I and $y - x < \delta$.

Proof. Choose a compact $[c, d] \subset I$ with $\int_{I - [c,d]} |f| < \epsilon/2$; f is bounded on $[c, d]$, say by K. With $\delta = \epsilon/(2K + 1)$, $\int_x^y |f| \leq \int_{[x,y] \cap [c,d]} |f| + \int_{I-[c,d]} |f| < K\delta + \epsilon/2 < \epsilon$. ■

7.4 Spaces of integrable functions

Motivation. Integral norms on functions; their incompleteness under Riemann integration motivates Chapter 10.

Proposition 7.11 (7.4.1)

For a normed space X the following are equivalent:
(a) every Cauchy sequence in X converges;
(b) whenever $\sum\|x_n\| < \infty$, the series $\sum x_n$ converges in X.

Proof. (a)⇒(b): the partial sums are Cauchy, since $\|\sum_{m<k\le n} x_k\| \le \sum_{m<k\le n}\|x_k\|$.

(b)⇒(a): given a Cauchy (y_n), choose $n_1 < n_2 < \cdots$ with $\|y_m - y_n\| < 2^{-k}$ for $m, n \ge n_k$. With $x_1 = y_{n_1}$, $x_{k+1} = y_{n_{k+1}} - y_{n_k}$, $\sum\|x_k\| < \infty$, so $y_{n_k} = \sum_{j\le k} x_j$ converges; a Cauchy sequence with a convergent subsequence converges (Lemma 2.13 of Volume I, same proof). ■

Let $\mathcal{R}^1(I)$ and $\mathcal{R}^2(I)$ be the locally Riemann integrable $f : I \to \mathbb{C}$ with $\int_I|f| < \infty$, resp. $\int_I|f|^2 < \infty$, with $\|f\|_1 = \int_I|f|$ and $\|f\|_2 = (\int_I|f|^2)^{1/2}$. These are seminorms ($\|f\| = 0 \not\Rightarrow f = 0$); $\mathcal{R}^2$ has inner product $\langle f, g\rangle = \int_I f\bar{g}$. Neither space is complete (hence Chapter 10). A **step function** is a finite linear combination of characteristic functions of bounded intervals.

Lemma 7.12 (7.4.2)

For $f \in \mathcal{R}^2(I)$ and $\epsilon > 0$ there is a step function s with $\|f - s\|_2 < \epsilon$.

Proof. Let f be real (else split). Choose $[a, b] \subset I$ with $\int_{I-[a,b]}|f|^2 < \epsilon^2/4$, $|f| \le K$ on $[a, b]$, a partition with $U(f, P) - L(f, P) < \epsilon^2/(8K)$, and $s = m_i$ on the i-th interval (0 off $[a, b]$). Then $0 \le f - s \le 2K$ on $[a, b]$, so $\int_a^b|f - s|^2 \le 2K\int_a^b(f - s) \le 2K(U - L) < \epsilon^2/4$. Altogether $\|f - s\|_2^2 < \epsilon^2/2$. ■

Lemma 7.13 (7.4.3)

For a step function s on I and $\epsilon > 0$ there is a continuous g with $\|s - g\|_2 < \epsilon$ (and $g = 0$ outside a compact subinterval).

Proof. Take $s = \chi_{[a,b]}$ (any end points) and $g = 1$ on $[a + \eta, b - \eta]$, 0 off (a, b), linear between; $|s - g| \le 1$ is nonzero only on two intervals of length η, so $\|s - g\|_2^2 \le 2\eta$. ■

Theorem 7.14 (7.4.4)

For $f \in \mathcal{R}^2(I)$ and $\epsilon > 0$ there is a continuous g with $\|f - g\|_2 < \epsilon$.

Proof. Combine Lemmas 7.12 and 7.13 with the triangle inequality for $\|\cdot\|_2$. ■

Corollary 7.15 (7.4.5)

For $f \in \mathcal{R}^2[a, b]$ there are polynomials P_n with $\|f - P_n\|_2 \to 0$.

Proof. Choose continuous g with $\|f - g\|_2 < \epsilon$ and a polynomial P with $\|g - P\|_{\sup} < \epsilon$ (Theorem 7.4); then $\|g - P\|_2 \le \epsilon\sqrt{b - a}$. ■

Theorem 7.16 (7.4.6)

For $f \in \mathcal{R}^1(I)$ and $\epsilon > 0$ there is a step function s with $\|f - s\|_1 < \epsilon$, and a continuous g vanishing outside a compact subinterval with $\|f - g\|_1 < \epsilon$. For $f \in \mathcal{R}^1[a, b]$ there are polynomials P with $\|f - P\|_1 < \epsilon$.

Proof. As in Lemma 7.12, with $\int_a^b |f-s| = \int_a^b (f-s) \le U - L$ for the lower step function. For the continuous approximation of $\chi_{[a,b]}$ the error in $\|\cdot\|_1$ is at most η. Polynomials: $\|g - P\|_1 \le (b-a)\|g-P\|_{\sup}$. ■

7.5 Selected exercises

Motivation. Book exercises: moments, Bernstein polynomials, and compactness in ℓ^2.

Exercise 7.5.1 (Moments determine a function)

Let $f : [0,1] \to \mathbb{R}$ be continuous with $\int_0^1 f(x) x^n \, dx = 0$ for $n = 0, 1, 2, \ldots$. Show that $f = 0$.

Solution.

Step 1. By linearity, $\int_0^1 fP = 0$ for every polynomial P.

Step 2. Take polynomials $P_n \to f$ uniformly (Theorem 7.4):

$$\int_0^1 f^2 = \int_0^1 f(f - P_n) \le \|f\|_{\sup} \|f - P_n\|_{\sup} \to 0,$$

so $\int_0^1 f^2 = 0$.

Step 3. If $f(x_0) \neq 0$, then $f^2 > \tfrac{1}{2} f(x_0)^2$ on an interval of length $\eta > 0$, so $\int_0^1 f^2 \ge \tfrac{1}{2} f(x_0)^2 \eta > 0$, a contradiction. ■

Exercise 7.5.5 (Weights concentrating at an end point)

Let $f : [0,1] \to \mathbb{R}$ be continuous. Show that

$$\lim_{n \to \infty} \frac{\int_0^1 x^n f(x) \, dx}{\int_0^1 x^n \, dx} = f(1).$$

Solution.

Step 1. As $\int_0^1 x^n dx = \frac{1}{n+1}$, the quotient minus $f(1)$ is $(n+1) \int_0^1 x^n (f(x) - f(1)) dx$.

Step 2. With $M = \max |f|$ and $|f(x) - f(1)| < \epsilon$ on $[1-\delta, 1]$,

$$(n+1) \int_0^{1-\delta} x^n |f(x) - f(1)| \, dx \le 2M(1-\delta)^{n+1},$$
$$(n+1) \int_{1-\delta}^1 x^n |f(x) - f(1)| \, dx \le \epsilon \, (n+1) \int_0^1 x^n dx = \epsilon.$$

Step 3. The first bound $\to 0$, so the quantity in Step 1 is $< 2\epsilon$ for large n. ■

Exercise 7.5.7 (Bernstein polynomials)

Let $f : [0,1] \to \mathbb{R}$ be continuous, $r_k(x) = \binom{n}{k} x^k (1-x)^{n-k}$, and $p_n(x) = \sum_{k=0}^n \binom{n}{k} f\left(\frac{k}{n}\right) x^k (1-x)^{n-k}$.
(a) Show that $\sum_{k=0}^n (k - nx)^2 r_k(x) = nx(1-x)$.
(b) Show that $p_n \to f$ uniformly on $[0,1]$.

Solution.

(a)

Step 1. By the binomial theorem, $\sum_k r_k(x) = (x + (1-x))^n = 1$.

Step 2. From $k\binom{n}{k} = n\binom{n-1}{k-1}$ and $k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}$, the binomial theorem again gives

$$\sum_k k\, r_k(x) = nx, \qquad \sum_k k(k-1)\, r_k(x) = n(n-1)x^2.$$

Step 3. Expanding $(k-nx)^2 = k(k-1) + k - 2nxk + n^2x^2$,

$$\sum_k (k-nx)^2 r_k(x) = n(n-1)x^2 + nx - 2n^2x^2 + n^2x^2 = nx(1-x).$$

(b)

Step 1. By (a), Step 1, $f(x) - p_n(x) = \sum_k (f(x) - f(\tfrac{k}{n}))r_k(x)$. Let $M = \max|f|$ and $|f(x)-f(y)| < \epsilon$ for $|x-y| < \delta$.

Step 2. Indices with $|\tfrac{k}{n} - x| < \delta$ contribute $\le \epsilon \sum_k r_k(x) = \epsilon$.

Step 3. The others have $(k-nx)^2 \ge n^2\delta^2$, so by (a) contribute at most

$$2M \sum_k \frac{(k-nx)^2}{n^2\delta^2} r_k(x) = \frac{2Mx(1-x)}{n\delta^2} \le \frac{M}{2n\delta^2}.$$

Step 4. So $\|f - p_n\|_{\sup} \le \epsilon + \frac{M}{2n\delta^2} < 2\epsilon$ for large n: a constructive proof of Theorem 7.4. ∎

Exercise 7.5.9 (Compact and non-compact boxes in ℓ^2)

Show that $K = \{a \in \ell^2 : |a(n)| \le \frac{1}{n},\ n = 1, 2, \dots\}$ is compact, i.e. every sequence in K has a subsequence converging in ℓ^2 to a point of K. Is $L = \{a \in \ell^2 : |a(n)| \le \frac{1}{\sqrt{n}},\ n = 1, 2, \dots\}$ compact?

Solution.

(a) K is compact.

Step 1. For (a_m) in K, $|a_m(n)| \le \frac{1}{n}$; the diagonal argument of Theorem 7.3 gives a subsequence (b_j) with $b_j(n) \to a(n)$ for all n.

Step 2. $|a(n)| \le \frac{1}{n}$, so $a \in K$.

Step 3. Choose N with $\sum_{n>N} \frac{4}{n^2} < \frac{\epsilon^2}{2}$ and j_0 with $\sum_{n \le N} |b_j(n) - a(n)|^2 < \frac{\epsilon^2}{2}$ $(j \ge j_0)$. As $|b_j(n) - a(n)| \le \frac{2}{n}$,

$$\|b_j - a\|_2^2 = \sum_{n \le N} |b_j(n) - a(n)|^2 + \sum_{n > N} |b_j(n) - a(n)|^2 < \epsilon^2 \qquad (j \ge j_0).$$

(b) L is not compact.

Step 1. For $m \ge 1$ let $c_m(n) = \frac{1}{\sqrt{n}}$ for $2^m < n \le 2^{m+1}$ and $c_m(n) = 0$ otherwise. Then $c_m \in L$ and

$$\|c_m\|_2^2 = \sum_{2^m < n \le 2^{m+1}} \frac{1}{n} \ge 2^m \cdot \frac{1}{2^{m+1}} = \frac{1}{2}.$$

Step 2. Disjoint supports give $\|c_m - c_{m'}\|_2^2 \ge 1$ $(m \ne m')$: no subsequence is Cauchy. ∎